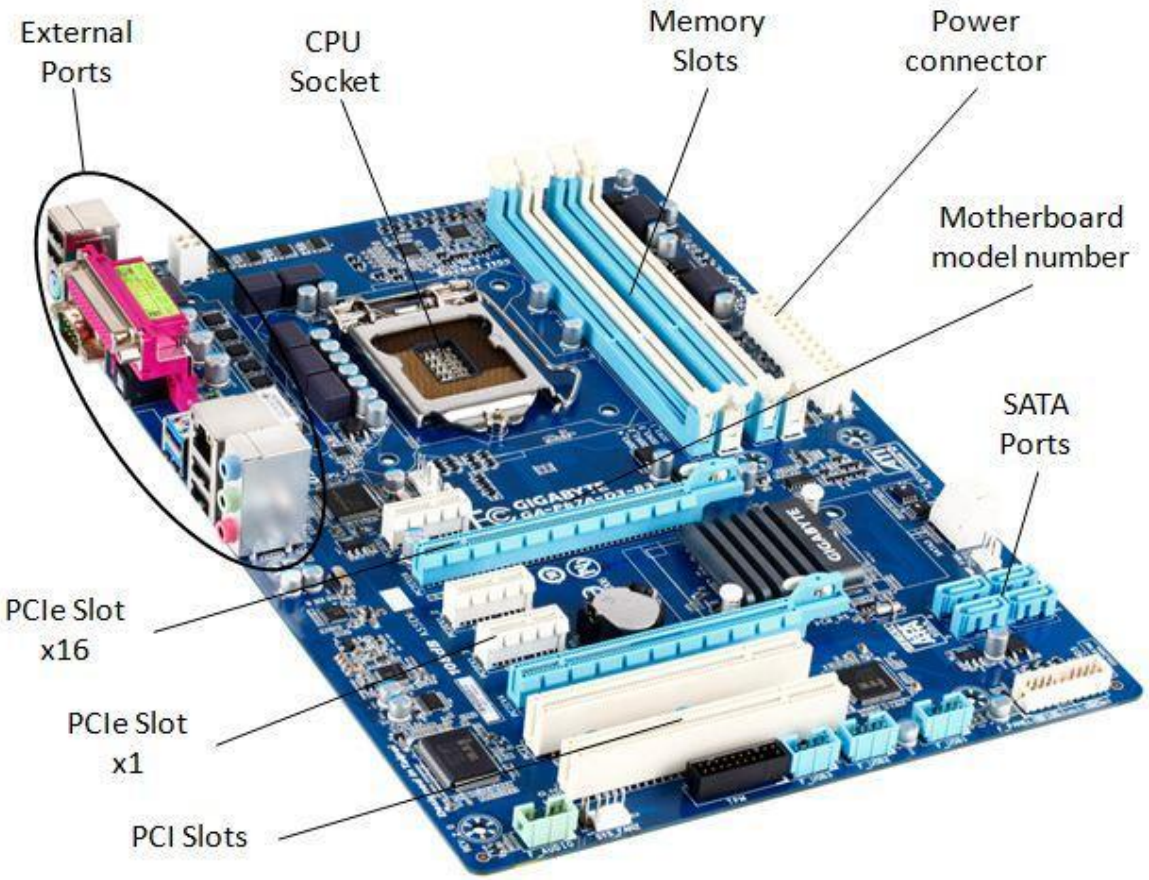
Motherboard and CPU

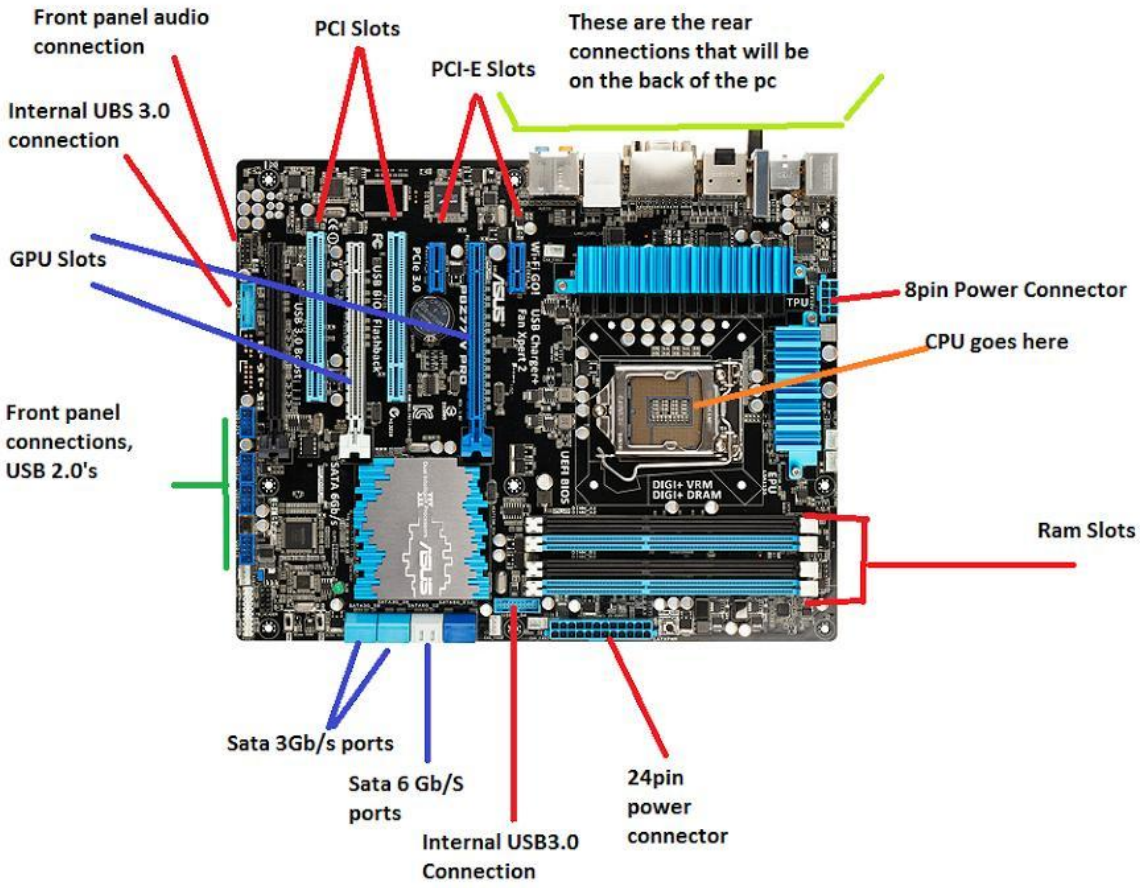
Q-1 Take a clear photo of the motherboard inside your own or a friend's laptop/desktop (or find a labeled image online), and label the CPU socket, RAM slots, PCIe slots, and power connectors using any image editing tool.

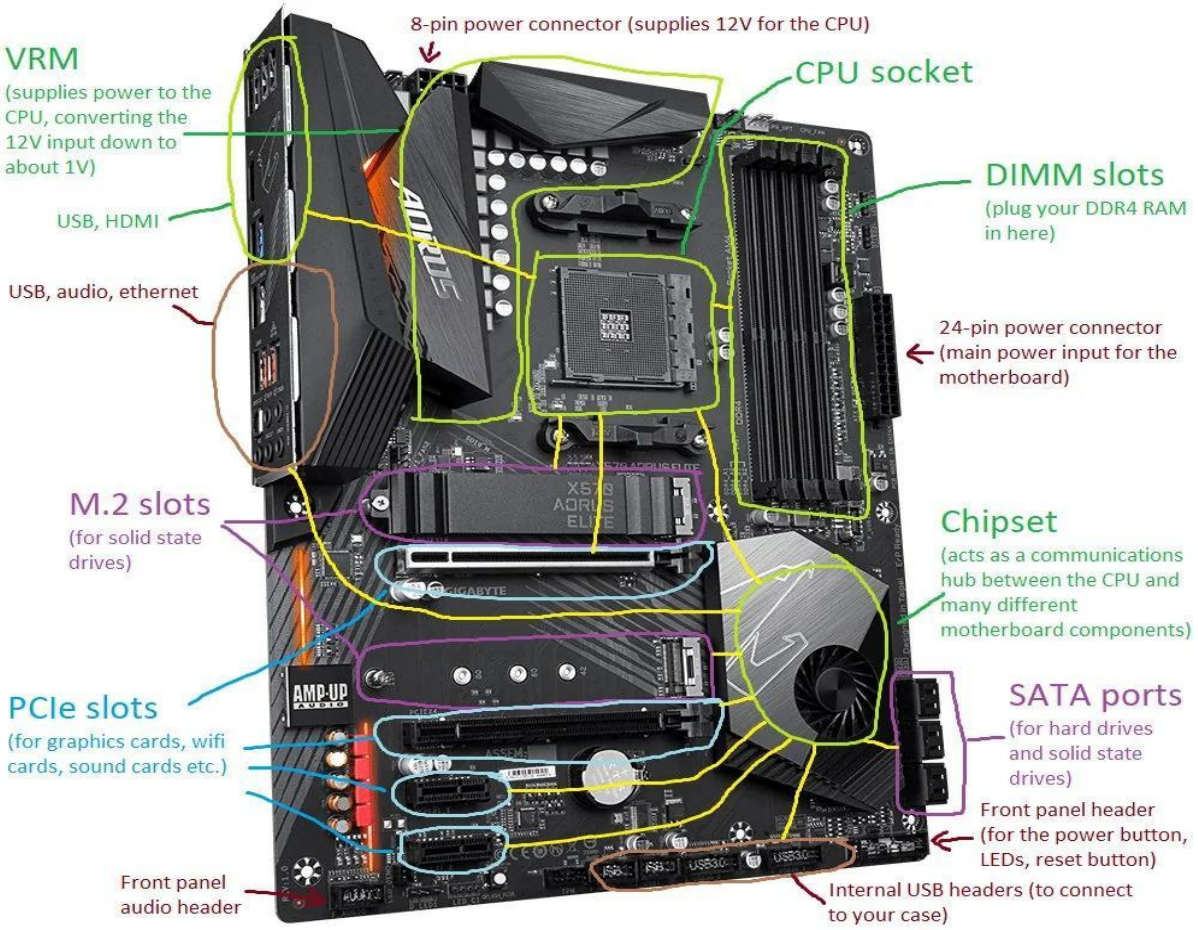
Ans: **Motherboard Component Labeling**

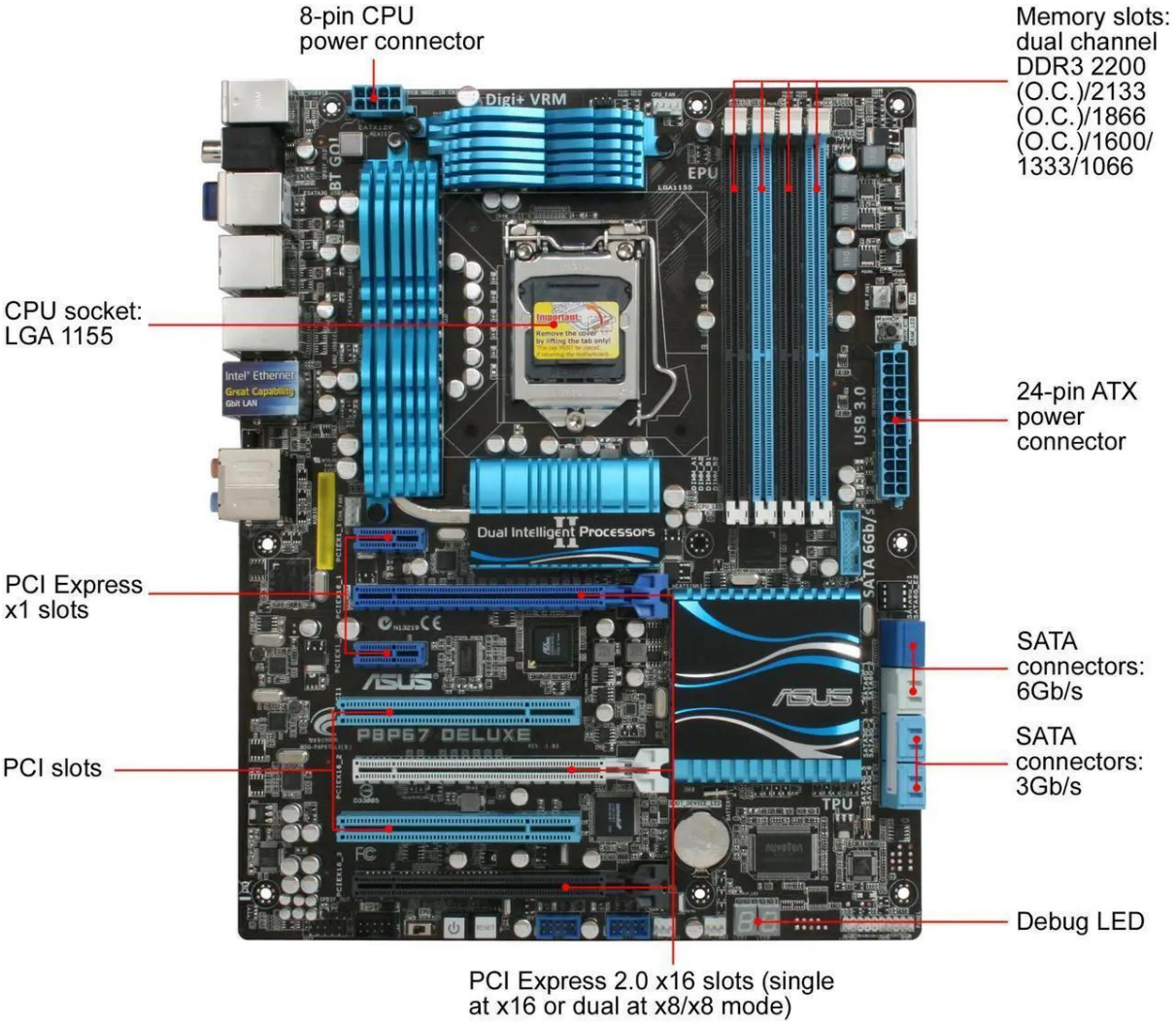
For this task, use a **clear motherboard image** and label these four parts with arrows:

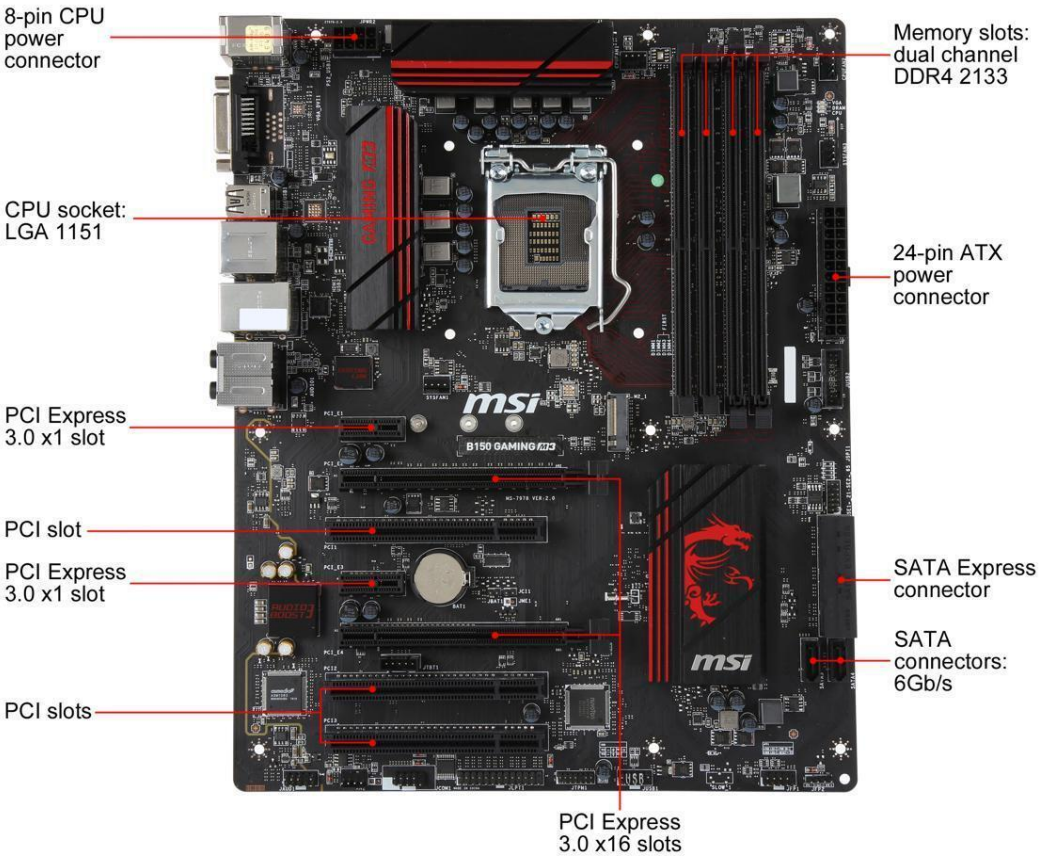
1. **CPU Socket** – The place where the CPU/processor is installed.
2. **RAM Slots (DIMM Slots)** – Slots where RAM sticks are installed.
3. **PCIe Slots** – Slots used for expansion cards such as a graphics card.
4. **Power Connectors** – Connectors that receive power from the power supply, such as the 24-pin motherboard connector and CPU power connector.

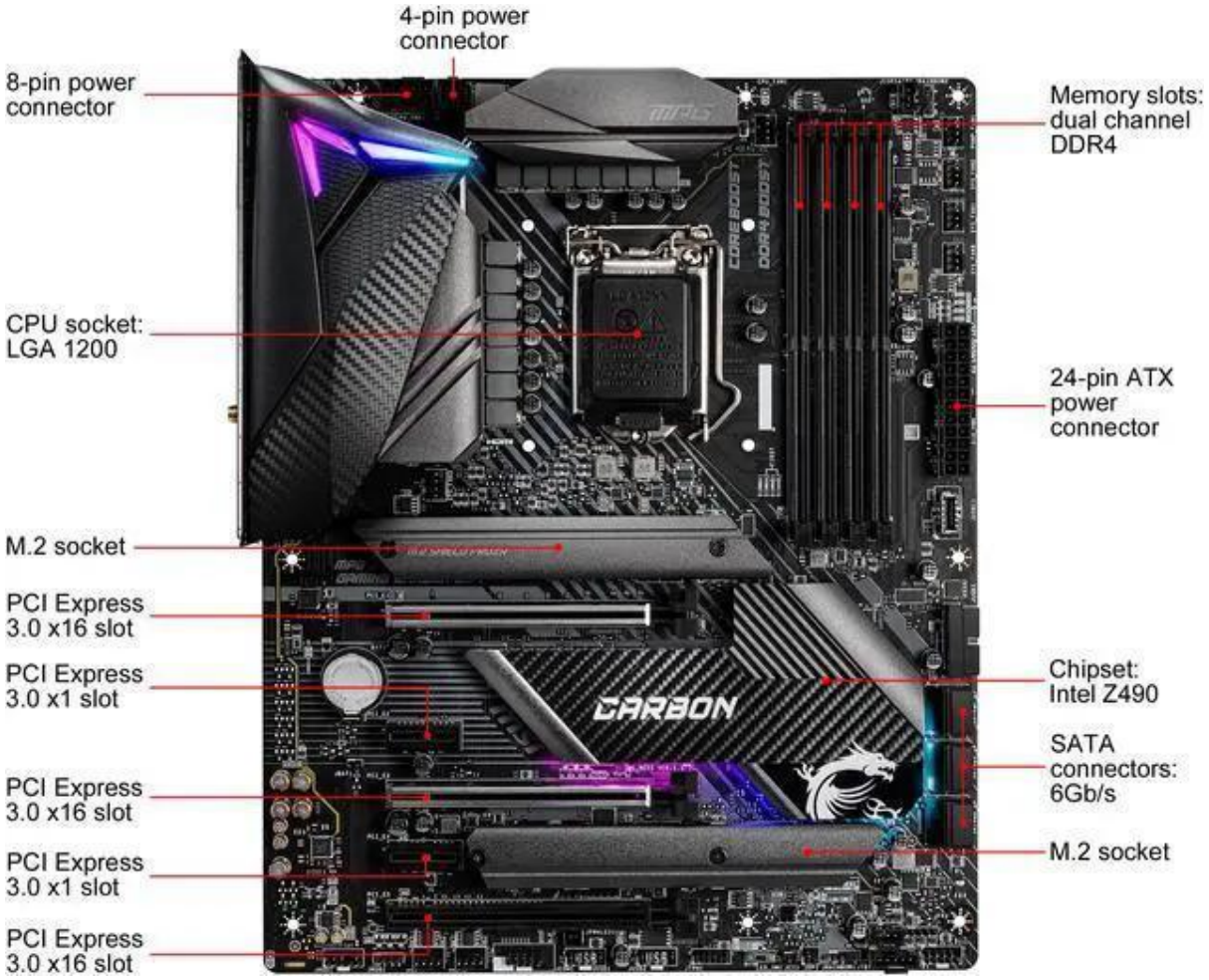



Front panel audio connection
PCI Slots
PCI-E Slots
These are the rear connections that will be on the back of the pc
Internal UBS 3.0 connection
GPU Slots
8pin Power Connector
CPU goes here
Front panel connections, USB 2.0's
Ram Slots
Sata 3Gb/s ports
Sata 6 Gb/S ports
Internal USB3.0 Connection
24pin power connector


8-pin power connector (supplies 12V for the CPU)
VRM
(supplies power to the CPU, converting the 12V input down to about 1V)
CPU socket
DIMM slots
(plug your DDR4 RAM in here)
USB, HDMI
USB, audio, ethernet
24-pin power connector (main power input for the motherboard)
M.2 slots
(for solid state drives)
Chipset
(acts as a communications hub between the CPU and many different motherboard components)
PCIe slots
(for graphics cards, wifi cards, sound cards etc.)
SATA ports
(for hard drives and solid state drives)
Front panel header
(for the power button, LEDs, reset button)
Front panel audio header
Internal USB headers (to connect to your case)


8-pin CPU power connector
Memory slots: dual channel DDR3 2200 (O.C.)/2133 (O.C.)/1866 (O.C.)/1600/ 1333/1066
CPU socket: LGA 1155
24-pin ATX power connector
PCI Express x1 slots
SATA connectors: 6Gb/s
SATA connectors: 3Gb/s
PCI slots
Debug LED
PCI Express 2.0 x16 slots (single at x16 or dual at x8/x8 mode)
Digi+ VRM
EPU
Dual Intelligent Processors
TPU
P8P67 DELUXE
ASUS


8-pin CPU power connector
CPU socket: LGA 1151
PCI Express 3.0 x1 slot
PCI slot
PCI Express 3.0 x1 slot
PCI slots
Memory slots: dual channel DDR4 2133
24-pin ATX power connector
SATA Express connector
SATA connectors: 6Gb/s
PCI Express 3.0 x16 slots
msi
B150 GAMING



Assignment format: Take one clear image → open it in **MS Paint / PowerPoint / Canva** → draw arrows → add the four labels above → save the edited image and attach it to your assignment.

Q-2 Pick any three popular CPUs used in gaming laptops or desktops (for example, Intel Core i5-12400F, AMD Ryzen 5 5600, Intel Core i7-13700K). Research and create a table comparing their clock speed (GHz), number of cores/threads, cache size (L2/L3), TDP (W), and current price in India.

Ans: **Comparison of Three Popular Gaming CPUs**

CPU	Clock Speed	Cores / Threads	Cache (L2 / L3)	TDP	Current Price in India
Intel Core i5-12400F	2.50 GHz base / up to 4.40 GHz	6 / 12	7.5 MB / 18 MB	65 W	₹15,500
AMD Ryzen 5 5600	3.50 GHz base / up to 4.40 GHz	6 / 12	3 MB / 32 MB	65 W	₹13,100
Intel Core i7-13700K	3.40 GHz P-core / up to 5.40 GHz	16 / 24	24 MB / 30 MB	125 W	₹33,985

Sources: Intel lists the i5-12400F at 6 cores/12 threads, 7.5 MB L2, 18 MB L3, and 65 W base power. AMD lists the Ryzen 5 5600 at 6 cores/12 threads, 3 MB L2, 32 MB L3, and 65 W TDP. Intel lists the i7-13700K at 16 cores/24 threads, 24 MB L2, 30 MB L3, and 125 W base power. Current Indian listings show ₹15,500 for the i5-12400F, ₹13,100 for the Ryzen 5 5600, and ₹33,985 for the i7-13700K.

Q-3 For each CPU in your comparison table, identify at least one compatible motherboard model (mention brand and model, e.g., ASUS PRIME B660M-A for Intel 12th Gen) and specify the socket type and form factor (ATX, mATX, ITX).

Ans: Compatible Motherboards for the Three CPUs

CPU	Compatible Motherboard	Brand & Model	Brand & Model	Form Factor
Intel Core i5-12400F	ASUS PRIME B660M-A D4	ASUS	LGA1700	mATX
AMD Ryzen 5 5600	MSI B550-A PRO	MSI	AM4	ATX
Intel Core i7-13700K	ASUS TUF GAMING Z790-PLUS WIFI	ASUS	LGA1700	ATX

Q-4 Visit the official website of any one CPU you compared (Intel or AMD), and find the maximum supported RAM speed and PCIe version for that CPU. Write these details along with how they affect real-world performance in gaming or multitasking.

Hint: Look for 'Memory Types' and 'PCI Express Revision' in the CPU specs page.

Ans: Intel Core i5-12400F – RAM and PCIe Specifications

Specification	Details
CPU	Intel Core i5-12400F
Maximum Supported RAM Speed	Up to **DDR5-4800 MT/s** or **DDR4-3200 MT/s**
PCI Express Version	PCIe 5.0 and PCIe 4.0
Maximum PCIe Lanes	20

Effect on Real-World Performance

RAM Speed:
Faster RAM helps the CPU access data more quickly. This can improve gaming performance, especially in CPU-heavy games, and helps when running multiple applications at the same time. The i5-12400F supports up to DDR5-4800 or DDR4-3200.

PCIe Version:
PCIe 5.0 provides a high-speed connection between the CPU and devices such as a graphics card or NVMe SSD. In gaming, a suitable PCIe connection helps the GPU and storage device transfer data quickly. In normal gaming, the actual performance also depends on the GPU, SSD, motherboard, and other hardware.

Conclusion:
The Intel Core i5-12400F supports modern high-speed RAM and PCIe 5.0/4.0, making it suitable for gaming and multitasking.